INCH-POUND

MIL-C-71189 (AR)
23 June 1994

MILITARY SPECIFICATION

CARTRIDGE, 81MM, INFRARED ILLUMINATING, XM816,
LOADING, ASSEMBLING, AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for the loading, assembling and packing for one type of mortar cartridge designated as Cartridge, 81MM, Infrared Illuminating, XM816.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

MIL-STD-286	-	Propellants, Solid, Sampling, Examination and Testing
MIL-STD-414	-	Sampling Procedures and Tables For Inspection by Variables for Percent Defective
MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9354333	-	Container, Ammunition, Fiber, PA114
9362799	-	Packing and Marking for Container, Ammunition Fiber for Cartridge, 81MM: M816, M819 and M853A1
12953389	-	Cartridge, 81MM, Infrared Illuminating, XM816 W/Fuze, MTSQ M772
12944509	-	Packing and Marking for Container, Ammunition, PA156 and PA157; for Cartridge 81MM
12944510	-	Container, Ammunition, Metal, PA157

INSPECTION EQUIPMENT DRAWINGS

6518071	-	Alignment Gage
6518072	-	Alignment Check Gage
7257965	-	Limit Check Gage
9200964	-	Photometer Head With Cell
9201136	-	Tunnel, Light
9201268	-	Procedure, Light Output Measurement
9201390	-	Procedure, Photocell Checkout
9247071	-	Photocell

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

(Copies of other Government documents, drawing, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12953389, all associated drawings, and with all requirements specified in applicable specifications and drawings.

3.3 Moisture.

3.3.1 First fire composition. The moisture content of the first fire composition shall not exceed 0.30 percent, when tested as specified in 4.4.5.3.1 or 4.4.5.3.3.

3.3.2 Illuminant composition. The moisture content of the illuminant composition shall not exceed 0.30 percent, when tested as specified in 4.4.5.3.2 or 4.4.5.3.3.

3.4 Static functioning of the illuminant assembly. The static burn time shall be not less than 80 seconds excluding first fire. The average infrared energy output for the first 80 seconds must exceed 75 watts/steradian in the near IR spectrum, of wavelengths between 0.60 to 0.95 micrometers. Determination shall be as specified in 4.4.5.4 (see 6.7). The average visible light output shall be less than 700 candela.

3.5 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.4.5.6.

3.5.1 Misfire. The cartridge shall not fail to fire from the weapon.

3.5.2 Fuze separation. The fuze shall function and the tail cone shall separate from the body tube assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

3.5.3 Functioning. The cartridge shall function satisfactorily (see 6.4) and shall provide effective illumination (see 6.5) for 60 seconds, min.

3.5.4 Security of parts. There shall be no metal parts failure in the bore of the weapon or in flight.

3.5.5 Horizontal range. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.5.6 Fuze function time. The fuze assembly shall function and shall comply with the fuze function time in Table I.

TABLE I. Fuze Function Time

Fuze setting (seconds)	Charge	Fuze function time (seconds)
13.5	1	13.5 $\pm$ 1.0
33.5	4	33.5 $\pm$ 1.0

3.5.7 Premature burst. The fuze shall not function in the bore of the weapon or in the first 100 meters of flight.

3.6 Pull test for oval sleeve crimp to suspension wire. The oval sleeve crimp to the suspension wire shall withstand a pull force of 1800 pounds minimum.

3.7 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.8 Workmanship.

3.8.1 Parts and assemblies. All parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion, and other foreign material that may adversely affect fit or function. The cleaning method used shall not be injurious to the parts nor shall any parts be contaminated.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Cloth. All cloth shall be free of tears, rips, and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears, or foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

3.8.4 Painting and marking. The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed per Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, 81MM: Infrared Illuminating XM816 Assembly	1 OF 1	See below
			NEXT HIGHER ASSEMBLY: N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	First Fire Composition (dwg. 12953387)	4 oz.	3.3.1	4.4.3.1/4.4.5.3.1/ 4.4.5.3.3
	Illuminant Composition (dwg. 12953387)	4 oz.	3.3.2	4.4.3.1/4.4.5.3.2/ 4.4.5.3.3
	Static Functioning (dwg. 12953387)	10	3.4	4.4.3.2/4.4.5.4
	Projectile Loading Assembly (dwg. 12953388)			
	Examination for defects	50 1/	3.2	4.4.2.4/4.4.2.5
	Cartridge (dwg. 12953389)			
	Examination for defects	176 1/	3.2	4.4.2.6/4.4.2.7
	Ballistic Test (Proving Ground)	176 1/	3.5	4.4.3.3/4.4.5.6

NOTES:
1/ Same group of samples

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see Table II) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot of Cartridges, 81MM, Infrared Illuminating shall contain:

a. Projectile body metal parts in accordance with the same drawing, same drawing revision, same specification, and same specification revision, of one lot interfix number, from one manufacturer.

b. Ignition cartridges, of one lot interfix number, from one manufacturer.

c. Fin assemblies, of one lot interfix number, from one manufacturer.

d. Propelling charges, of one lot interfix number, from one manufacturer.

e. Loaded fuzes of one interfix from one manufacturer.

f. Parachute assemblies from one lot interfix number from one manufacturer.

g. Canister assemblies from one manufacturer, in accordance with the same drawing and same drawing revision.

h. Obturating rings, of one lot interfix, from one manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table III below, using the inspection levels stated in the Classification of Characteristics paragraphs.

Table III. Attributes sample inspection.

	Inspection Levels					
Lot Size	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.9). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Canister Illuminant Metal Parts Assembly	1 OF 1	9381084
			NEXT HIGHER ASSEMBLY: 12953387

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Point stake inadequate	Level V	3.2	Visual/Manual
202	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26 04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Canister Loading Assembly (Prior to Loading Illuminant Composition)	1 OF 1	12953387
			NEXT HIGHER ASSEMBLY: 12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Coating on inside surface of canister missing, inadequate or tacky	Level III	3.2	Visual/Manual
Minor 201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89

Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26104:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Canister Loading Assembly (After Loading Illuminant Composition)	1 OF 1	12953387
			NEXT HIGHER ASSEMBLY: 12953388

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ignition disc assembly missing	Level III	3.2	Visual
102	Filler rings not flush to 1/16 inch above flush	Level III	3.2	Gage
103	Swivel fails to rotate freely	100%	3.2	Visual/Manual
104	Static functioning test		3.4	4.4.3.2/4.4.5.4
105	First fire composition moisture content test		3.3.1	4.4.3.1.1/4.4.5.3.1
106	Illuminant composition moisture content test		3.3.2	4.4.3.1.2/4.4.5.3.2
Minor				
201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.4	Projectile, 81MM: Infrared Illuminating XM816 Loading Assembly During Assembly Operation	SHEET 1 OF 1	12953388
			NEXT HIGHER ASSEMBLY 12953389

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Shroud lines pulled out of parachute bag beyond max. drawing requirement	Level III	3.2	Gage
102	Parachute bag closure cord not removed	Level III	3.2	Visual
103	Pull cord improperly attached to parachute disc, parachute assembly or drogue plate	Level III	3.2	Visual
104	Pull test for oval sleeve to suspension wire	1/	3.2	4.4.5.10
105	Taped shroud loops not attached to suspension wire loop	Level III	3.2	Visual
106	Tape missing from the sleeve and cable	Level III	3.2	Visual
Minor				
201	Canister loading assembly improperly seated	Level V	3.2	Visual
202	Spacer felt misaligned or missing	Level V	3.2	Visual
203	Cable loop attachment not above separator disc	Level V	3.2	Visual
204	Separator disc improperly positioned or missing	Level V	3.2	Visual
205	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

1/ See paragraph 4.4.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Projectile, 81MM: Infrared Illuminating XM816 Loading Assembly Including Tail Cone Assembly	1 OF 2	12953388
			NEXT HIGHER ASSEMBLY: 12953389

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Warning missing, incomplete, or incorrect	100%	3.2	Visual
B	Obturator ring missing 1/	100%	3.2	Visual
Major				
101	Depth of rear shear pin holes, 2/	Level III	3.2	Gage
102	Location of rear shear pin holes from end of body tube	Level III	3.2	Gage
103	Diameter of rear shear pin holes, 2/	Level III	3.2	Gage
104	Shear pin(s), rear missing or loose	100%	3.2	Visual
105	Shear pin(s) not flush or below	100%	3.2	Gage
106	Protective dust plug missing (if applicable)	Level III	3.2	Visual
107	Entire obturator including tack weld not flush or below bourrelet diameter, 1/ and 3/	100%	3.2	Gage
108	Tack weld missing 1/	100%	3.2	Visual
109	Tail cone not fully seated	Level III	3.2	Gage

NOTES:

1/ If attachment of the obturator ring is done during cartridge assembly, these defects will be inspected at that time.

2/ Alternate sampling plan may be used as specified in paragraph 4.4.5.9.

3/ The inspection shall be performed at two places for each obturator, one at the tackweld and the other at 180 degrees from the tackweld.

4/ If a total of two or more defective units are found, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.5	Projectile, 81MM: Infrared Illuminating XM816 Loading Assembly Including Tail Cone Assembly	SHEET 2 OF 2	12953388
			NEXT HIGHER ASSEMBLY 12953389

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
110	Pull cord pinched between drogue plate and split sleeves	Level III	3.2	Visual
111	Excess silastic present on OD of body tube or tail cone	Level III	3.2	Visual
112	Tail cone improperly assembled to the body tube assembly 4/	Level III	3.2	4.4.5.11/Visual
Minor				
201	Markings incorrect, missing or illegible	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cartridge, 81MM Infrared Illuminating XM816 w/Fuze MTSQ, M772 (Prior to Assembly of Fuze Fin Assembly & Ignition Cartridge	1 OF 1	12953389

NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Damaged threads on fuze	Level III	3.2	Visual
102	Damaged threads on ignition cartridge	Level III	3.2	Visual
103	Damaged threads on cartridge housing	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Cartridge, 81MM Infrared Illuminating XM816 W/Fuze MTSQ, M772	1 OF 2	12953389
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Ignition cartridge missing, cracked or damaged	100%	3.2	Visual
B	Propelling charge increment(s) missing, cracked or damaged	100%	3.2	Visual
C	Fuze safety wire missing	100%	3.2	Visual
D	Fuze not set on safe	100%	3.2	Visual
Major				
101	Assembly fails to freely traverse alignment gage for 20 inches, min.	100%	3.2	Gage
102	Inadequate torque applied to fin boom	Level III	3.2	Torque test/4.4.5.5
103	Inadequate torque applied to fuze	Level III	3.2	Torque test/4.4.5.5
104	Damaged fin blades	Level III	3.2	Visual/Magnifier
105	Security of parts	Level III	3.2	Manual/4.4.5.7
106	Total length, max.	Level III	3.2	Gage
107	Inadequate torque applied to ignition cartridge	Level III	3.2	Torque test/4.4.5.5
108	Identifying strip missing	100%	3.2	Visual
Minor				
201	Protective finish damaged	Level V	3.2	Visual
202	Markings incorrect, illegible or missing	Level V	3.2	Visual
203	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assistdla.mil -- Downloaded: 2026-10-26 04:37Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Cartridge, 81MM Infrared Illuminating XM816 W/Fuze MTSQ, M772	2 OF 2	12953389
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	PROVING GROUND TEST			
	Ballistic Acceptance	96/50	3.5	4.4.3.3/4.4.5.6

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Container, Ammunition, Fiber, for Cartridges XM816/M819/M853A1/	1 OF 1	9362799
			NEXT HIGHER ASSEMBLY: 12944509

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Parts not secure to cap or container	Level III	3.2	Manual/4.4.5.7
102	Container damaged	Level III	3.2	Visual
103	Absence of gap between cap and tube prior to taping	Level III	3.2	Visual/5.1
104	Increment tape missing	Level III	3.2	Visual
105	Propelling charge support missing or damaged	Level III	3.2	Visual
106	Tab at the end of increment tape inadequate	Level III	3.2	Visual
107	Tab at the end of increment tape covered by propellant charge support label	Level III	3.2	Visual
Minor				
201	Marking incorrect, illegible or missing	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Container, Ammunition, Fiber, for Cartridges XM816/M819/M853A1 After Securing	1 OF 1	9362799
			NEXT HIGHER ASSEMBLY: 12944509

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Cartridge loose inside container	Level III	3.2	Manual
102	Inner sealing strip not applied smoothly or oriented improperly	Level III	3.2	Visual
103	Outer sealing strip not applied smoothly or oriented improperly	Level III	3.2	Visual
104	Outer sealing strip not folded into tab at end	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Packing and Marking for Container, Ammo. PA156 & PA157 for Cartridge, 81MM (Prior to Closing)	1 OF 1	12944509
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Containers loose or improperly packed in metal can	Level III	3.2	Visual/Manual
102	Support assembly damaged, missing, or improperly placed	Level III	3.2	Visual/Manual
Minor				
201	Circular firing scale not present in every third metal box	Level V	3.2	Visual
202	Ammo box containing circular firing scale not appropriately marked	Level V	3.2	Visual
203	Plastic sheet not present in every sixth box	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-71189 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Packing and Marking for Container, Ammo. PA156 & PA157 for Cartridge, 81MM (After Closing)	1 OF 1	12944509
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Contents of box loose	Level III	3.2	Manual
102	Cover assembly not secure	Level III	3.2	Manual
Minor				
201	Marking missing, misleading or unidentifiable	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.3.1 Moisture content.

4.4.3.1.1 First fire composition. The contractor shall provide adequate controls to ensure compliance with the requirements and shall select a three to four ounce first fire sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.1.2 Illuminant composition. The contractor shall provide adequate controls to ensure compliance with the requirements and shall select a three to four ounce illuminant sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.2 Static functioning. The test shall be performed as specified in 4.4.5.4. A sample for the First Article Inspection is specified in Table II. A sample for the inspection lot shall be selected in accordance with the sampling plan shown in 4.4.3.2.a. The candle loading assemblies shall be tested for static burning time and candle output. If the average static burning time is less than 80 seconds excluding first fire or the average infrared energy output in the first 80 seconds is less than 75 watts/steradian, the inspection lots represented by the samples shall be rejected. No individual test sample shall have the candle output less than 70 watts/steradian minimum in the first 80 seconds or the average visible light output is more than 700 candela, otherwise the inspection lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

a. A sample of three candles from each inspection lot (approximately 85 candles) shall be selected. The candle assemblies shall be tested for compliance with the requirements specified in 4.4.3.2. If the sample of three candles fails to comply with the requirements specified in 4.4.3.2, a second sample of three candles shall be tested. If the second sample of three candles also fails to comply with the requirements specified in 4.4.3.2, the inspection lot represented by the sample shall be rejected. This sampling frequency shall be continued until 15 consecutive lots (15 groups of 3 candles) comply with the specified requirements.

b. After successful completion of 4.4.3.2.a, the sampling frequency may be reduced to three candles from each group of six inspection lots (approximately 510 candles). The canister assemblies shall be tested for compliance with the requirements in 4.4.3.2. If the sample of three candles from each group of six inspection lots fails to comply with the requirements specified in 4.4.3.2, a second sample of three candles shall be tested. If the second sample of three candles also fails to comply with the requirement specified in 4.4.3.2, the group of six inspection lots shall be rejected. This sampling frequency shall be continued until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.2.a.

c. After successful completion of 4.4.3.2.b, the sampling frequency may be reduced to three candles from each group of fifteen inspection lots (approximately 1275 candles). The canister assemblies shall be observed for compliance with the requirements specified in 4.4.3.2. If the sample of three candles from each group of fifteen inspection lots fails to comply the requirements specified in 4.4.3.2, a second sample of three candles shall be tested. If the second sample of three candles also fails to comply the requirements specified in 4.4.3.2, the group of fifteen inspection lots shall be rejected. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.2.b.

d. In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.4.3.2.b and in the event of a production stoppage greater than two weeks, the sampling plan shall revert to 4.4.3.2.a. If the mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall be reverted to 4.4.3.2.b.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

e. In the event of failure of any sample to comply with requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot.

4.4.3.3 Proving ground.

a. First article test. One hundred and seventy-six (176) sample cartridges shall be selected for first article testing. The cartridges shall be separated into three groups; one group (48 samples) conditioned for 24 hours at plus 145 ± 2.5° F, one group (80 samples) conditioned for 24 hours at plus 70 ± 2.5° F, and one group (48 samples) conditioned for 24 hours at minus 50 ± 2.5° F. Each group shall be separated into equal subgroups; one subgroup tested at charge 1 and one subgroup tested at charge 4.

b. Initial production test. Beginning with the first lot produced and continuing until three consecutive lots have successfully passed the acceptance criteria specified herein, ninety-six (96) sample cartridges shall be selected for this test. The cartridges shall be separated into three groups: one group (34 samples) conditioned for 24 hours at plus 145 ± 2.5° F, one group (28 samples) conditioned for 24 hours at plus 70 ± 2.5° F, and one group (34 samples) conditioned for 24 hours at minus 50 ± 2.5° F. Each group shall be separated into equal subgroups; one subgroup tested at charge 1 and one subgroup tested at charge 4.

c. Production test. After successfully completing the initial production test, the next ten consecutive production lots (50 samples) shall be conditioned for 24 hours at plus 70 ± 2.5° F. The next production lot (eleventh lot) shall be tested and evaluated in accordance with the initial production test method of paragraph 4.4.3.3.b (96 samples). Upon successful completion of the eleventh lot, the next production lot shall be counted as the beginning of another ten consecutive lots (50 samples). Again, the eleventh production lot shall be tested and evaluated in accordance with the initial production test method (96 samples) and so on. Upon unsuccessful completion of any eleventh lot, the ballistic test shall revert to the requirements of paragraph 4.4.3.3.b.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.3.3.1 Individual acceptance criteria. The test shall be performed as specified in Defect classification and accept/reject criteria shall be in accordance with the following table:

Categories	Defect	Accept/Reject	
Critical:	Premature burst or metal parts separation	0	1
Major:	Misfire	1	2
Major:	Fuze non-function or non-separation of tail cone from body tube assembly	1	2
Major:	Horizontal range less than 80% of average range	0	1
Major:	Fuze functioning time (See Note 1) See para. 4.4.3.3.2.1		
Major:	Illuminant fails to ignite or provide effective illumination for less than 60 seconds in the air (See Note 2 & 3)	2	3

Note 1: If the fuze performance of any round in the sample exceeds the timing requirements specified, and the round functions properly in all respects with the exception that the "effective illumination" time is less than sixty seconds because the illuminant and parachute assembly hit the ground while still burning, the illumination time defects shall not be additionally assessed against that round.

Note 2: If the fuze performance of any round in the sample meets the fuze functioning time requirements within plus or minus one second of the fuze settings specified, the mortar elevation (QE) is proper and the "effective illumination" time is less than sixty seconds, the performance of the cartridge will be considered valid for evaluating the acceptability of the lot for illumination time defects. The following are considered some causes of this defect:

a. Streaming chute/ripped chute.
b. Partial or fully tangled chute.
c. Chute fails to deploy
d. Delayed deployment of the parachute.
e. Illuminant detached from canister (illuminant dropout).
f. Parachute detached from canister.

Note 3: Fuze functioning time is defined as the elapsed time from round initiation in the mortar tube to first flash as fuze ejects the payload.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.3.3.2 Overall acceptance criteria. In addition to the individual accept/reject criteria of 4.4.3.3.1, the production lot sample shall be rejected if a total of three or more major defects occur. The initial production lots and any eleventh lot (96 samples) shall be rejected if a total of four or more major defects occur. In the event that neither a misfire nor a fuze non-function/non-separation defect occurs during testing, the initial production and any eleventh lot allows a maximum of three (3) major defects for illumination less than 60 seconds in the air. The first article sample shall be rejected if a total of six or more major defects occur. In the event that neither a misfire nor a fuze non-function/non-separation defect occurs during testing, the first article allows a maximum of five major defects for illumination less than 60 seconds in the air. However, if either one misfire or one non-function/non-separation defect occurs, the first article allows a maximum of four major defects for illumination less than 60 seconds in the air. If one misfire and one non-function/non-separation defect occur, the first article allows a maximum of three (3) major defects for illumination less than 60 seconds. If either two or more misfires or non-function/non-separation defects occur during testing, the first article shall be rejected.

4.4.3.3.2.1 Acceptance criteria for fuze. The lot shall also be rejected if the percent lot defective for fuze functioning time as computed with the use of MIL-STD-414, Section B, with an AQL of 1.5 percent, exceeds the maximum allowable percent lot defective for each setting.

4.4.3.4 Strength of heat weld. One obturating ring sample, minimum, after every 100 projectiles, shall be assembled to a fixture, introduced to the production line, heat sealed and tested as specified in 4.4.5.8. Failure of the sample to meet the requirement shall be cause for those 100 preceding projectiles represented by the sample to be rebanded and retested.

4.4.3.5 Pull test for oval sleeve crimps to suspension wire. (See drawing 9209555-8, 9381075 & 12953388) Pull test shall be conducted in accordance with paragraph 4.4.5.10.

4.4.3.5.1 Sampling testing plan. One assembly shall be selected at the start of each 4 hours of each production from each crimping machine to qualify the machine. A sample shall also be selected immediately following any adjustment to the machine which effects the crimping load regardless of when the adjustment is made. The test shall be performed as specified in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.5.10. Failure of the sample to withstand a 1800 pound minimum pull force shall be cause for rejection of previous 4 hours production made on the applicable crimping machine.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.4.5 Methods of inspection.

4.4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing, or specification requirements, and that all specified manufacturing processes have been followed.

4.4.5.2 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3, 4, and 5 of the specification, in accordance with 4.4.2 herein.

4.4.5.3 Moisture.

4.4.5.3.1 First fire composition. The moisture content of the first fire composition shall be determined in accordance with MIL-STD-1234, Method 101.3.1. The special solvent to be used shall be equal parts by volume of dimethylformamide and anhydrous methanol.

4.4.5.3.2 Illuminant composition. The moisture content of illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in MIL-STD-1234, Method 101.2.1 up to paragraph 5.3. A 50 ml. sample of 1:1 dimethylformamide methyl alcohol solution shall be added into a 250ml. Erlenmeyer flask (fitted with a rubber stopper) containing 10 grams of the pyrotechnic sample. (Exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix the solution and pyrotechnic sample. Allow the solution to set 5 to 15 minutes. A blank sample of 1:1 dimethylformamide methanol solution shall be prepared as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

described above. A 60 ml. portion of methanol shall be transferred through the sample porthole into the dry titration beaker containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for at least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary end point is reached. All succeeding operations shall be completed within 3.50 - 0.25 minutes. Twenty-five (25) ml. of the supernatant solution shall be rapidly introduced, accurately pipetted from the supernatant liquid into the beaker through the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 mls) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation: $\frac{AB}{5\ (10)}$ =

	Percent H_2O
Less	Std Blank
Leaves	Percent Moisture in Sample

Where:

A = mls of Karl Fischer to titrate sample
B = Standardization Factor (mg./ml.)
10 = Constant Factor
5 = Standard Sample Weight

4.4.5.3.3 Alternate method for first fire composition moisture and illuminant composition moisture. The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4, MIL-STD-286. The temperature setting shall be 100° C and the time setting 30 minutes. The sample size shall be between 0.40 and 0.60 grams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.5.4 Static functioning test.

4.4.5.4.1 Effective radiance and burn time. The test equipment required for this test shall be submitted for Government approval (see 6.3). The temperature of candle shall be 70° ± 5° F at time of test. The average effective radiance is to be calculated, using the first 80 seconds of burn time, by dividing the total integral (watt-sec/steradian) over this period by 80. The first fire disc assemblies shall be removed prior to testing and the candles shall be burned at upward position. The distance from the radiometric cell to the flare shell shall be 30.00 ± .25 feet. The axial center of the radiometric cell and the axial center of the fireball shall be located 54.0 ± .5 inch from the floor. The temporary back wall listed on dwg. 9201136 shall be removed so that the distance from the center of the candle to the back wall is 14-15 feet. The center line of baffle opening shall be in line with axial center of fireball and axial center of radiometric cell. The test shall be conducted in a hearth that has been painted flat black. Depending on ash buildup the tunnel hearth shall be repainted at least once in a three month duration to maintain the black interior or more frequently as determined by the resident Government Quality Assurance Representative. If the average burning time or average effective radiance does not meet the requirement of paragraph 4.4.3.2 the illuminant inspection lot represented by the sample shall be rejected.

4.4.5.4.2 Average visible light output. The test equipment required for this test is delineated in paragraph 2.1.2 under Inspection Equipment Drawings. The average candlepower is to be calculated, using the first 80 seconds of burn time, by dividing the total integral (candela-seconds) over this period by 80. The distance from the photometric cell to the flare shell shall be 30.00 ± .25 feet. The axial center of the photometric cell and the axial center of the fireball shall be located 54.0 ± .5 inch from the floor. The center line of baffle opening shall be in line with axial center of fireball and axial center of photometer cell. If the average visible light output exceeds the requirement of paragraph 4.4.3.2, the illuminant inspection lot represented by the sample shall be rejected.

4.4.5.5 Torque test. The samples supplied in accordance with 4.4.2.7 shall be tested by applying the minimum required torque to the components in the direction of assembly before the thread sealing compound has set. Any perceptible movement of parts will be considered as failure of the assembly to comply with the requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.5.6 Proving ground. The test shall be performed at a Government Proving Ground. All the cartridges shall be fired from an 81mm, M252 Mortar for which the round is standard. The cartridge shall be assembled with accepted fuzes, fin assemblies, adapters, ignition cartridges, and propellant increments. One half of the sample cartridges shall be fired with one propelling charge increment (charge zone one) and one half of the sample cartridges shall be fired with four propelling charge increments (charge zone four) and in accordance with Table III. However, if meteorological conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to increase the height of burst and insure effective illumination. Observation shall be made for compliance with the requirements specified for security of parts, functioning, functioning time, horizontal range and "effective illumination" (See 6.5). Fuze functioning time shall be reported to at least the nearest 0.10 sec, illuminant burn time to the nearest 0.5 second, and horizontal range on the ground to a point under the position of fuze functioning to the nearest 5 meters. The cartridges shall be conditioned at the assigned temperatures for a minimum of 24 hours prior to testing. Any malfunction occurring during the test shall be reported in the official firing record. Meteorological data shall also be recorded in the official firing record. Muzzle velocity and chamber pressure shall also be recorded for each round fired and reported in the official firing record.

TABLE III Proving ground test

Charge Zone	Fuze Setting (seconds)	Mortar Elevation (degrees)
1	13.5	45
4	33.5	45

4.4.5.7 Security of parts. Parts to be inspected for security shall be manually checked to be tightly and securely attached as described in the applicable drawings. Such parts will not be readily loosened or evidence rattling or other apparent signs of poor assembly when so checked.

4.4.5.8 Strength of heat weld. The sample shall be tested for integrity by applying the required load at the heat weld radially and in a direction away from the axis (center) of the ring.

4.4.5.9 Alternate sampling plan for examination of depth and diameter of rear shear pin holes. The depth and diameter of the rear shear pin holes shall be inspected in the following manner using approved gages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.5.9.1 First production shift examination. Beginning with the first production shift, ten (10) consecutive drilled units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every 1/2 hour until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in any of the five (5) units, the machine set up shall be adjusted until ten (10) consecutive units are defect free.

4.4.5.9.2 Second production shift examination. After successfully completing the sampling of paragraph 4.4.5.9.1 for one eight hour shift of production with no defects, the sampling plan may be reduced to five (5) drilled units being inspected every one hour and at the beginning and end of the shift. If a defect is found, the sampling plan of paragraph 4.4.5.9.1 shall be reinstated.

4.4.5.9.3 Third production shift examination. After successfully completing the sampling plan of paragraph 4.4.5.9.2 for one eight hour production shift with no defects, the sampling plan may be reduced to three (3) drilled units being inspected every hour and at the beginning and end of each shift. If a defect is found, the sampling plan of paragraph 4.4.5.9.2 shall be reinstated.

4.4.5.9.4 Fourth production shift examination. After successfully completing the sampling plan of paragraph 4.4.5.9.3 for one eight hour shift with no defects, the sampling plan may be reduced to two (2) drilled units being inspected every two hours. If a defect is found, the sampling plan of 4.4.5.9.3 will be reinstated.

4.4.5.9.5 Tooling change. A change in tooling will require a first piece inspection prior to continuing production. The sampling plan will be the same as that prior to the tooling change.

4.4.5.9.6 Production lapse. A stop in production of over two months will require starting the inspection at the first production shift plan. See 4.4.5.9.1.

4.4.5.10 Pull test for oval sleeve to suspension wire. The test sample shall be assembled on a universal testing machine and the load applied in an axial direction until the required pull force is exerted. The rate of travel of the machine shall be 2 inches per minute maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

4.4.5.11 Tail cone and body tube inspection. At the end of each assembly lot produced, fifty random samples shall be selected for this inspection. The projectile loading assemblies shall be subjected to a pull load sufficient to shear the pins; slowly remove the body tube assembly from the tail cone. Inspect the pull cord for the following: Pull cord should be coiled about one half of the length and inserted into a 3/4 inch loop attaching the pull cord to the parachute bag and the other one half should be between the parachute disc and the droque plate. If the pull cord does not meet the above or is tangled or pinched, the sample shall be classed as defective. In addition, if the cable is frayed, kinked or deformed, the sample shall be classed as defective.

5. PACKAGING

5.1 Unit packaging.

5.1.1 Level A. Packaging for cartridges shall be in accordance with the requirements of drawing 9362799.

5.1.2 Level B. Same as level A.

5.1.3 Level C. Not applicable.

5.2 Packing.

5.2.1 Level A. Cartridges shall be packed in accordance with the requirements of drawing 12944509.

5.2.2 Level B. Same as level A.

5.2.3 Level C. Not applicable

5.3 Marking.

5.3.1 Level A. Packaged cartridges shall be marked in accordance with the requirements of drawing 9362799. Packed cartridges shall be marked in accordance with requirements of drawing 12944509.

5.3.2 Level B. Same as level A.

5.3.3 Level C. Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for assembly to XM816 Infrared illuminating cartridge for use on 81mm mortar.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Satisfactory functioning. Satisfactory functioning shall be defined as ejection of the parachute and illuminant assembly and proper suspension and ignition of the illuminant such as to assure "effective illumination".

6.5 Effective illumination. Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

6.6 Illuminant inspection lot. Illuminant from the mixer will be collected in containers of approximately equal quantity of illuminant which will be numbered and used in sequence. Illuminant candle assemblies will be assigned an inspection lot number for each 85 candles produced. Smaller lot sizes will be accepted when dictated by interruption or shutdown of operations.

6.7 Static burning time. Minimum static burning time as referenced herein (see 3.4) is required, as a time base for the determination of luminosity and compliance does not in itself guarantee meeting the final ballistic minimum burning time requirement. Structural integrity of the candle under set back loading is of prime importance in meeting the ballistic burning time requirement and is attained by proper formulation blending, consolidation and cure.

6.8 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-B and SMCAR-FSA-MM, Picatinny Arsenal, NJ 07806-5000.

6.9 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.10 Subject term (key word) listing.
Cartridge
Functioning
Fuze
Horizontal range
Illuminant
Infrared
Premature
Security
Separation
Static

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AF55)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-71189 (AR)	940623

3. DOCUMENT TITLE: CARTRIDGE, 81MM, INFRARED ILLUMINATING, XM816, LOADING, ASSEMBLING, AND PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:31Z
Check the source to verify that this is the current version before use.